Physiology Practice Questions: DM Module

Lecture Title: Enteric Motility (L22)

Note: If you need any clarification, please email to: ASK-PHY@sgu.edu or email to the faculty who delivered this lecture.

1. A 40-year-old man comes to the physician because of a 3-month history of upper gastrointestinal discomfort and dysphagia. He says that he has a diffuse aching in his chest. Physical examination shows no abnormalities. Radiography shows distension of the lower esophagus caused by a failure of relaxation of the lower esophageal sphincter (LES). Manometry shows increased pressure in the LES during swallowing. Which of the following is the most likely diagnosis?

A. Gastroesophageal reflux
B. Barrett's esophagus
C. Gastritis
D. Achalasia
E. Peptic ulcer disease

2. A 30-year-old man is admitted to the hospital because of a motor vehicle collision. He is conscious but unable to move his lower limbs. Physical examination shows loss of sensations, weakness, and reflexes in both lower limbs. He has lost voluntary control of his bladder and bowel. Which of the following structures involved in defecation most likely is under voluntary control in a normal person?

A. Rectum
B. External anal sphincter
C. Internal anal sphincter
D. Longitudinal muscle in sigmoid colon
E. Circular muscle in sigmoid colon

3. A 7-month-old boy is brought to the physician by his mother because of reduced bowel movements since birth. The mother says that he passed his first meconium 4 days after birth, and since then, has been defecating once weekly with great difficulty. Physical examination shows a distended abdomen with palpable stool in the lower quadrants. Histological examination of a biopsy of the distal colon shows absence of myenteric plexus. Which of the following gastrointestinal dysfunctions is most likely in this patient?

A. Sluggish peristalsis in the segment of the gastrointestinal tract involved
B. Hypersecretion of acid by the stomach
C. Hypersecretion of the electrolytes and water by the small intestine
D. Indigestion of fatty foods
E. Increased pressure in the lower esophageal sphincter

4. A 24-year-old man participates in a study related to the functions of various sphincters in the gastrointestinal tract. Contraction and relaxation of sphincters are estimated by recording pressure changes in these sphincters. Increase in pressure indicates contraction and decrease in pressure indicates relaxation of these sphincters. Pressure recordings are done during various procedures like swallowing, distension of stomach, distension of ileum, distension of colon, and injection of various hormones. Which of the following procedures most likely increase pressure in the ileocecal sphincter in this participant?

A. Swallowing a meal
B. Distention of the ileum
C. Distention of the colon
D. Distention of stomach
E. Injection of CCK

5. A 26-year-old man comes to the physician for a routine examination. He says that he can eat his favorite food even after he feels full after a heavy meal. He has no history of any illness. Vital signs are within normal limits. Physical examination shows no abnormalities. The physician reassures the patient and tells him that this is normal because of the great capacity of the stomach to temporarily store food. This function of the stomach is called receptive relaxation. Which of the following most likely mediates this function of the stomach?

A. Smooth muscles in antral region of the stomach
B. Vagus nerve
C. Swallowing center in the medulla
D. Glossopharyngeal nerve
E. Gastrin

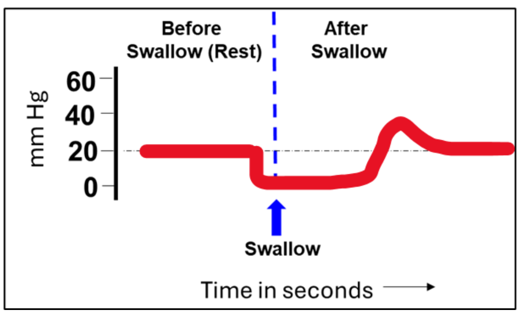
6. Investigators conduct a study relating to the effect of various factors on the small intestinal motility. Investigators use pig’s small intestine in vivo, for this purpose. Placing meat pieces in the proximal ileum initiates a propulsive movement called peristalsis. Which of the following best characterizes this type of intestinal movement?

A. Mixes the food bolus
B. It stops by cutting extrinsic nerves
C. Involves simultaneous contraction of smooth muscles behind and in front of the food bolus
D. Involves contraction of smooth muscles behind the food bolus and relaxation of smooth muscles in front of the bolus
E. Involves simultaneous relaxation of smooth muscles behind and in front of the bolus

7. A 34-year-old man comes to the physician for a routine examination. He says his appetite is normal but feels full for a longer time after eating food rich in fats. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following mechanisms best explains this patient's symptom?

A. Reduced rate of gastric emptying by fatty food
B. Stimulation of vagus after eating
C. Increased rate of gastric emptying by fatty food
D. Increased H+ secretion by fatty food
E. Increased mixing and grinding movements by fatty food

8. A 36-year-old man comes to the physician because of a 5-month history of difficulty swallowing and diffuse chest discomfort during eating. Physical examination shows no abnormalities. Barium esophagram shows a dilated esophagus. Histological examination of a biopsy from the lower esophagus shows absence of myenteric neural plexus. Esophageal manometry is performed, and the lower esophageal sphincter pressure is recorded. Which of the following LES pressure tracings is most likely in this patient?



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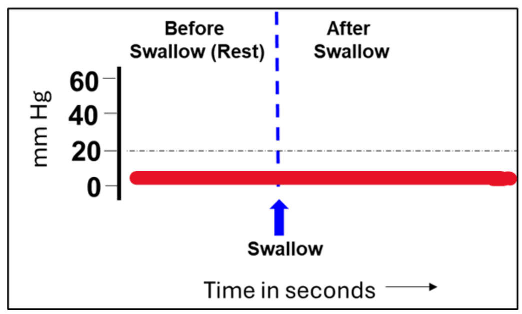

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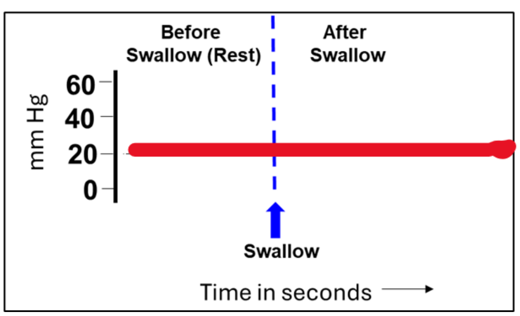

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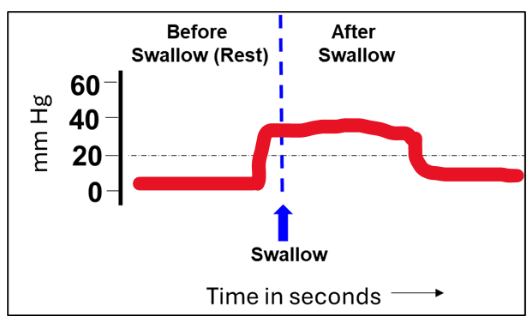

D

9. A 47-year-old woman comes to the physician because of a 10-month history of reduced frequency of passing stools which is hard in consistency. She says she drinks less fluid, and her diet contains less fibers. She has no history of chronic illness. Vital signs are within normal limits. Physical examination shows palpable mass in the left lower quadrant of the abdomen. Colonoscopy shows no structural abnormalities. She is advised to drink more fluids and add more fiber in her diet. Which of the following best characterizes motility in the colon in a normal person?

A. Increase In frequency in response to increased circulating levels of epinephrine
B. Haustral movements occur more often than peristaltic contractions
C. Occur at a higher frequency in the sigmoid colon than the transverse colon
D. Transit time is 2-3 hours
E. Mass movements help in water absorption in colon

10. A 77-year-old man comes to the physician because of a 2-month history of progressive weakness and abdominal discomfort. His vital signs are within normal limits. Physical examination shows pale conjunctivae, tongue, and nailbeds. Abdominal examination shows a palpable mass in the lower epigastric region. Upper GI endoscopy shows a tumor in the body of the stomach near the greater curvature. The mass is surgically removed along with the removal of middle third of the stomach, sparing fundal and antral regions. Which of the following motor functions of the stomach is most likely reduced after surgery in this patient?

A. Receptive relaxation
B. Retropulsion and grinding
C. Pyloric sphincter function
D. Intrinsic factor secretion
E. Beginning of peristaltic waves

Answer Key and Explanation

1. A 40-year-old man comes to the physician because of a 3-month history of upper gastrointestinal discomfort and dysphagia. He says that he has a diffuse aching in his chest. Physical examination shows no abnormalities. Radiography shows distension of the lower esophagus caused by a failure of relaxation of the lower esophageal sphincter (LES). Manometry shows increased pressure in the LES during swallowing. Which of the following is the most likely diagnosis?

A. Gastroesophageal reflux
B. Barrett's esophagus
C. Gastritis
D. Achalasia
E. Peptic ulcer disease

Answer Key: D
Rationale: Dysphagia means ‘difficulty swallowing’ which is a prominent symptom in this patient. Radiography and manometry findings suggest that this is specifically due to LES dysfunction. This disorder is called achalasia, because of dysfunction of myenteric plexus in the lower esophagus. This is a primary esophageal motility disorder characterized by the absence of esophageal peristalsis and impaired relaxation of the LES.

2. A 30-year-old man is admitted to the hospital because of a motor vehicle collision. He is conscious but unable to move his lower limbs. Physical examination shows loss of sensations, weakness, and reflexes in both lower limbs. He has lost voluntary control of his bladder and bowel. Which of the following structures involved in defecation most likely is under voluntary control in a normal person?

A. Rectum
B. External anal sphincter
C. Internal anal sphincter
D. Longitudinal muscle in sigmoid colon
E. Circular muscle in sigmoid colon

Answer Key: B
Rationale: During defecation, distention of the rectum evokes an urge to defecate. Parasympathetic nerves stimulate the enteric nervous system to contract rectum and relax the internal anal sphincter. These are involuntary, mediated through the pelvic nerve. At this stage of rectal distention, voluntary contraction of the external anal sphincter and the puborectalis muscle prevents leakage. Thus, decision to defecate is voluntary, which results in commands from the brain to the sacral spinal cord to stop excitatory input to the external anal sphincter and pelvic floor muscles. When convenient, external anal sphincter and pelvic floor muscles relax voluntarily along with contraction of the rectum that results in expulsion of the feces.

3. A 7-month-old boy is brought to the physician by his mother because of reduced bowel movements since birth. The mother says that he passed his first meconium 4 days after birth, and since then, has been defecating once weekly with great difficulty. Physical examination shows a distended abdomen with palpable stool in the lower quadrants. Histological examination of a biopsy of the distal colon shows absence of myenteric plexus. Which of the following gastrointestinal dysfunctions is most likely in this patient?

A. Sluggish peristalsis in the segment of the gastrointestinal tract involved
B. Hypersecretion of acid by the stomach
C. Hypersecretion of the electrolytes and water by the small intestine
D. Indigestion of fatty foods
E. Increased pressure in the lower esophageal sphincter

Answer Key: A
Rationale: This patient has congenital absence of myenteric plexus (Hirschsprung's disease). The myenteric plexus is the key regulator of the longitudinal and circular smooth muscle responsible for peristaltic motility. If it is insufficiently developed, the affected region would exhibit significantly reduced motility, as in this patient. Stomach acid secretion is not controlled by the myenteric plexus, neither is the digestion of fatty foods. Occasional bowel movements result from secretion of electrolytes and water, which is also not controlled by the myenteric plexus.

4. A 24-year-old man participates in a study related to the functions of various sphincters in the gastrointestinal tract. Contraction and relaxation of sphincters are estimated by recording pressure changes in these sphincters. Increase in pressure indicates contraction and decrease in pressure indicates relaxation of these sphincters. Pressure recordings are done during various procedures like swallowing, distension of stomach, distension of ileum, distension of colon, and injection of various hormones. Which of the following procedures most likely increase pressure in the ileocecal sphincter in this participant?

A. Swallowing a meal
B. Distention of the ileum
C. Distention of the colon
D. Distention of stomach
E. Injection of CCK

Answer Key: C
Rationale: In general, stretching GI structures upstream (proximal) of a sphincter tends to relax, whereas stretching downstream (distal to a sphincter) causes contraction. The ileocecal sphincter separates the small and large intestines (between ileum and cecum). The colon is downstream of the ileocecal sphincter; hence distension of the colon causes constriction (indicated by increased pressure) of the ileocecal sphincter. CCK has no direct effect on this sphincter.

5. A 26-year-old man comes to the physician for a routine examination. He says that he can eat his favorite food even after he feels full after a heavy meal. He has no history of any illness. Vital signs are within normal limits. Physical examination shows no abnormalities. The physician reassures the patient and tells him that this is normal because of the great capacity of the stomach to temporarily store food. This function of the stomach is called receptive relaxation. Which of the following most likely mediates this function of the stomach?

A. Smooth muscles in antral region of the stomach
B. Vagus nerve
C. Swallowing center in the medulla
D. Glossopharyngeal nerve

E. Gastrin

Answer Key: B
Rationale: Receptive relaxation occurs in the fundus and upper part of the body of the stomach (together called proximal stomach). This is by a vago-vagal reflex in which vagus nerve carries both ascending and descending signals. The entry of food into stomach stimulates the stretch receptors and the afferent signals from these receptors are carried through the vagus to the brain. The efferent signals are transmitted through the vagus nerve to the enteric nervous system (ENS). Increased activity of the inhibitory motor neurons in the ENS relaxes smooth muscles in this region of the stomach and produces receptive relaxation. Hence stomach can temporarily store food as a reservoir. This function is lost by vagotomy.

6. Investigators conduct a study relating to the effect of various factors on the small intestinal motility. Investigators use pig’s small intestine in vivo, for this purpose. Placing meat pieces in the proximal ileum initiates a propulsive movement called peristalsis. Which of the following best characterizes this type of intestinal movement?

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D. Involves contraction of smooth muscles behind the food bolus and relaxation of smooth muscles in front of the bolus
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Answer Key: D
Rationale: Peristalsis is a propulsive movement that moves bolus down in the GI tract (towards anus) by the combined action of smooth muscle in front of and behind the bolus of food. Mixing movements do not have a forward motion component and are called segmentation. Peristalsis is coordinated by the enteric nervous system and does not need extrinsic nerves. However, the extrinsic nerves modulates the amplitude (strength) of this motility.

7. A 34-year-old man comes to the physician for a routine examination. He says his appetite is normal but feels full for a longer time after eating food rich in fats. His vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following mechanisms best explains this patient’s symptom?

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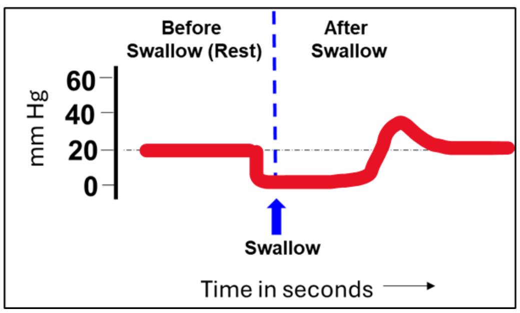
C. Increased rate of gastric emptying by fatty food
D. Increased H^+ secretion by fatty food
E. Increased mixing and grinding movements by fatty food

Answer Key: A
Rationale: Gastric emptying is slowed by the presence of luminal acidity and digestive substrates (especially amino acids, peptides, fatty acids) in the chyme arriving in the duodenum. These substrates stimulate release of CCK from I cells of the small intestine. CCK inhibits smooth muscles of the stomach and therefore reduces gastric motility. Reduced motility reduces antral pump mechanism and slows gastric emptying.

8. A 36-year-old man comes to the physician because of a 5-month history of difficulty swallowing and diffuse chest discomfort during eating. Physical examination shows no abnormalities. Barium esophagram shows a dilated esophagus. Histological examination of a biopsy from the lower esophagus shows absence of myenteric neural plexus. Esophageal manometry is performed, and the lower esophageal sphincter pressure is recorded. Which of the following LES pressure tracings is most likely in this patient?



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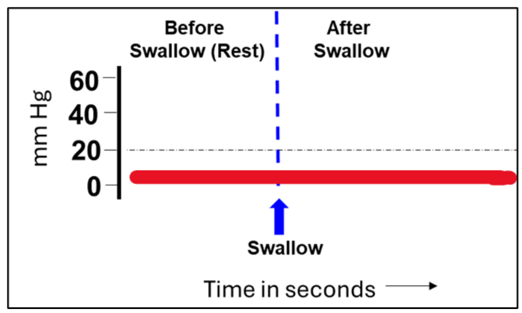

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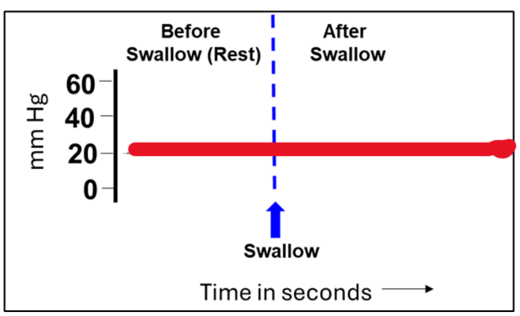

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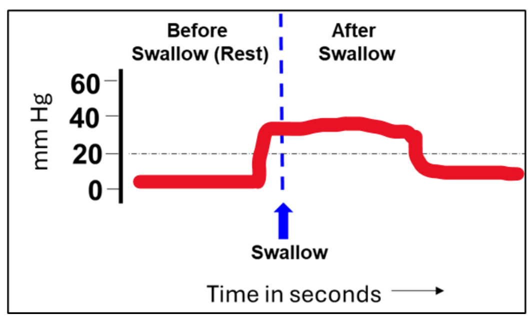

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Answer Key: C
Rationale: This patient has achalasia. In this condition, there is loss of peristalsis in the lower esophagus and LES fails to relax during swallowing. Therefore, LES pressure continues to be high as in rest (figure C). The defect is because of the absence of myenteric neural plexus in the esophagus. Figure A represents normal LES pressure changes during swallowing in a normal person.

9. A 47-year-old woman comes to the physician because of a 10-month history of reduced frequency of passing stools which is hard in consistency. She says she drinks less fluid, and her diet contains less fibers. She has no history of chronic illness. Vital signs are within normal limits. Physical examination shows palpable mass in the left lower quadrant of the abdomen. Colonoscopy shows no structural abnormalities. She is advised to drink more fluids and add more fiber in her diet. Which of the following best characterizes motility in the colon in a normal person?

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C. Occur at a higher frequency in the sigmoid colon than the transverse colon
D. Transit time is 2-3 hours
E. Mass movements help in water absorption in colon

Answer Key: B
Rationale: Haustral movements are a modified type of segmentation movement that occurs at a slow rate. These are the ringlike contractions of the circular muscle that divide the colon into pockets called haustra. Haustral formations and segments on either side of them remain in their respective states for long periods of time. This facilitates the absorption of water in the colon, without any net forward propulsion. Rate of colonic movements are very slow and therefore, it takes 8 to 15 hours to move the chyme from ileocecal valve through the colon. Mass movement is the second important movement in the colon. Mass movement is a modified peristalsis in which a segment (about 10-20 cm length) of colon loaded with fecal matter contracts as a single unit, propelling the fecal material in this segment “en masse” further down the colon. This movement facilitates movement of the fecal matter towards rectum and initiates defecation reflex.

10. A 77-year-old man comes to the physician because of a 2-month history of progressive weakness and abdominal discomfort. His vital signs are within normal limits. Physical examination shows pale conjunctivae, tongue, and nailbeds. Abdominal examination shows a palpable mass in the lower epigastric region. Upper GI endoscopy shows a tumor in the body of the stomach near the greater curvature. The mass is surgically removed along with the removal of middle third of the stomach, sparing fundal and antral regions. Which of the following motor functions of the stomach is most likely reduced after surgery in this patient?

A. Receptive relaxation
B. Retropulsion and grinding
C. Pyloric sphincter function
D. Intrinsic factor secretion
E. Beginning of peristaltic waves

Answer Key: E

Rationale: The peristaltic contractions begin in the body of the stomach. These contractions continue distally through the antrum, with the force of the contraction increasing downstream. Tonic and phasic contractions in the antral region cause mixing, grinding and retropulsion of food to reduce its particle size. Food particles that are less than 2 mm in size are propelled through the pyloric sphincter into the duodenum for further digestion and absorption.